# Proposal: AI powered experiments on $SL_2(\mathbb{Z}_p)$

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$SL_2(\mathbb{Z}_p)$ has two generators, $S$ and $T$. Given any $\gamma$ in $SL_2(\mathbb{Z}_p)$ we can consider how many applications of $S$ and $T$ are needed to reach it. There is a smallest such number, call it $\ell(\gamma)$, and as $SL_2(\mathbb{Z}_p)$ is finite, there exists some $\gamma$ in the group such that $\ell(\gamma)$ is greater than or equal to any other $\ell(\alpha), \alpha \in SL_2(\mathbb{Z}_p)$. This value is known as the diameter of the group, or informally, God's number. Denote God's number by $G(SL_2(\mathbb{Z}_p))$.

It is known that $G(SL_2(\mathbb{Z}_p)) = O(\log(p))$, but the constant is not known. Also, God's numbers for $SL_2(\mathbb{Z}_p)$ are not usually computed or tabulated. By computing $G(SL_2(\mathbb{Z}_p))$ for many small $p$, we find an experimental estimate for C.

Cayley graphs for $\mathrm{SL}_2(Z_p) equipped with the standard generators are ramanujan graphs, which are known for t$